Positive-GCD Q: Pair Bounds, Positive Definiteness, and a Zero-Diagonal Kill

Closure note — Theorems A/B/C proved; C′ false

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Operator. $Q_{ij} = \gcd(i,j)/\sqrt{ij}$ (positive GCD — not inverse-GCD $\widetilde{Q}$). No Navier–Stokes claim.

Claim	Status
A: $R(e_a - e_b) > 0$ on Q	**Proved**
B: $R_{\widehat{Q}}(e_a - e_b) \geq -1$	**Proved**
C: $Q_N \succ 0$ (all N)	**Proved**
C′: $\lambda_{\min}(\widehat{Q}) > -1/2$ for all N	**False**

Abstract

On the positive-GCD Gram matrix $Q_N(i,j) = \gcd(i,j)/\sqrt{ij}$ we prove restricted pair bounds (Theorems A–B), positive definiteness of Q_N for every N (Theorem C, via congruence with the classical GCD matrix), and kill the zero-diagonal full-spectrum claim $\lambda_{\min}(\widehat{Q}) > -1/2$. Companion inverse-GCD Bridge* lives in `04_q6_inverse_gcd.tex`; do not mix the two operators.

1 Operator

$$Q_N(i,j) = \frac{\gcd(i,j)}{\sqrt{ij}}, \qquad 1 \leq i,j \leq N.$$

In particular $Q_N(a,a) = 1$. Write $\widehat{Q}_N$ for Q_N with diagonal zeroed. Let $G_N(i,j) = \gcd(i,j)$ and $D = \operatorname{diag}(1,2,\ldots,N)$, so

$$Q_N = D^{-1/2}\, G_N\, D^{-1/2}.$$

2 Theorem A — diagonal pair positivity

Theorem 2.1 (A). *For integers $a \neq b$,*

$$R(e_a - e_b) = 1 - \frac{\gcd(a,b)}{\sqrt{ab}} \; > \; 0.$$

Proof. Set $g = \gcd(a,b)/\sqrt{ab}$. Then $g < 1$ for $a \neq b$, and $R = 1 - g > 0$. $\square$

3 Theorem B — zero-diagonal pair bound

Proposition 3.1 (B)**.** *For $a \neq b$,*

$$R_{\widehat{Q}}(e_a - e_b) = -\frac{\gcd(a,b)}{\sqrt{ab}} \ \geq \ -1.$$

Proof. Immediate from $\gcd(a,b)/\sqrt{ab} \leq 1$. □

4 Theorem C — positive definiteness

Theorem 4.1 (C)**.** *For every $N \geq 1$, the matrix Q_N is positive definite.*

Proof. The classical GCD matrix $G_N = (\gcd(i,j))_{1 \leq i,j \leq N}$ is positive definite (Beslin–Ligh; equivalently $\det G_N = \prod_{k=1}^{N} \varphi(k) > 0$ by Smith's formula, and the same for principal minors on initial segments). The diagonal matrix $D^{-1/2}$ is invertible, so $Q_N = D^{-1/2} G_N D^{-1/2}$ is congruent to G_N and hence positive definite. □

5 Kill C′ — zero-diagonal full spectrum

Theorem 5.1 (C′, kill)**.** *There exist N such that $\lambda_{\min}(\widehat{Q}_N) < -1/2$.*

Numeric certificate. E.g. $\lambda_{\min}(\widehat{Q}_5) \approx -0.793 < -1/2$ (`scripts/positive_gcd_floor_verify.py`). □

Remark 5.2 (Hygiene)**.** *Inverse-GCD $\widetilde{Q}$ already has $\lambda_{\min} < -1/2$ for moderate N. That kill does* not *transfer to positive-GCD Q (which is PD). Zero-diagonalization destroys the PD property and produces the C' counterexamples.*

References

[1] S. Beslin and S. Ligh, *Greatest common divisor matrices*, Linear Algebra Appl. **118** (1989), 69–76.

[2] H.J.S. Smith, *On the value of a certain arithmetical determinant*, Proc. London Math. Soc. **7** (1876), 208–212.

[3] J.R. Simons, *GCD Spectral Paper 1* (v1), Zenodo 10.5281/zenodo.20269738.